



Silicon Tracker Simulation and Performance Analysis using Geant4

From multiple scattering and resolution limits to secondary vertex and mass reconstruction

Nicolò Favagrossa

October 5, 2026



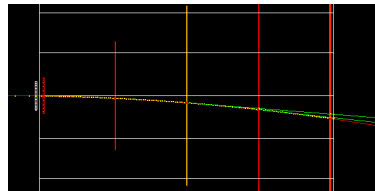
UNIVERSITÀ
DEGLI STUDI
DI MILANO



Experimental Setup: The SiTracker Simulation

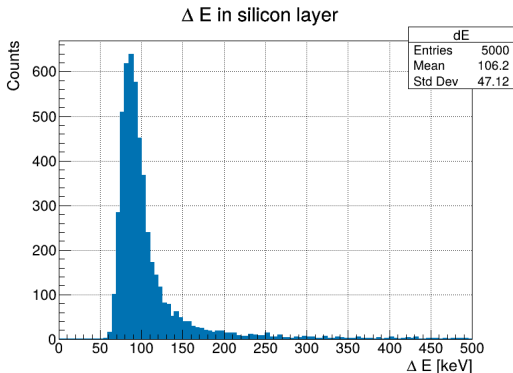
Based on the Geant4 exampleB2, this setup simulates a transverse slice of a generic collider tracker.

- **Beam Pipe:** 1 Beryllium layer (2 mm thick) emulating the vacuum chamber interface.
- **Tracking System:** 5 Silicon detector layers (300 μm thick) supported by Carbon structures (1 mm thick).
- **Magnetic Field:** Uniform field aligned along the x-axis (nominal $B = 1 \text{ T}$).





Exercise 1.1: Energy loss in silicon



- Particle: μ^- at $p = 100 \text{ GeV}/c$ (MIP).
- Target: $300 \mu\text{m}$ Silicon layer.
- Calculated expected loss:

$$\Delta E_{\text{exp}} = \left\langle \frac{dE}{dx\rho} \right\rangle \cdot x\rho = 116.2 \text{ keV}$$

- Extracted mean from dE histogram:

$$\Delta E_{\text{sim}} = 106.2 \text{ keV}$$

- The relative difference: $\sim 9.4\%$



Exercise 1.2: Multiple scattering in the beam pipe

Before reaching the first sensitive layer, a 100 GeV/c muon experiences multiple scattering primarily inside the 2 mm Beryllium beam pipe.

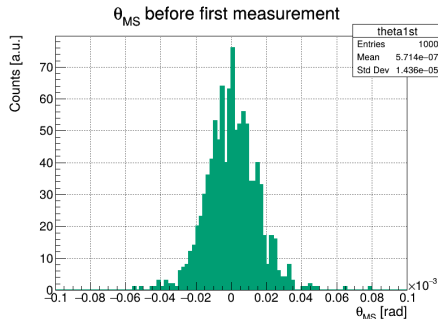
Using the parameterized formula:

$$\theta_0 = \frac{13.6 \text{ MeV}}{\beta c p} z \sqrt{\frac{x}{X_0}} \left[1 + 0.038 \ln \left(\frac{x}{X_0} \right) \right]$$

With $X_0(\text{Be}) \approx 35.3 \text{ cm}$, we obtain:

$$\theta_{\text{Be}} \approx 8.22 \times 10^{-6} \text{ rad}$$

However, the theta1st histogram yields a wider distribution: $\theta_{\text{MS}} \approx 1.44 \times 10^{-5} \text{ rad}$.



theta1st includes both the Be pipe **and** the 1st tracker layer (Si+C). Summing in quadrature with the layer's theoretical θ_{MS} yields the simulated value:

$$\theta_{\text{tot}} = \sqrt{\theta_{\text{Be}}^2 + \theta_{\text{layer}}^2} \approx 1.28 \times 10^{-5} \text{ rad}$$



Exercise 1.3: Multiple scattering in a tracker layer

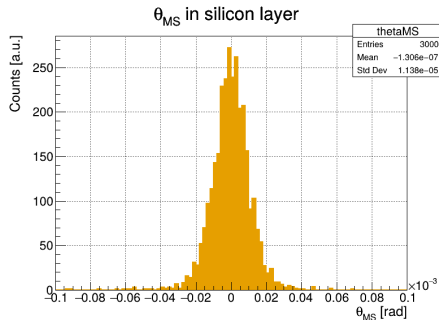
A single tracking layer consists of the active 300 μm Silicon sensor backed by a 1 mm Carbon support ($\rho = 2 \text{ g/cm}^3$).

To apply the parameterized formula to this composite material, we sum the fractional radiation lengths:

$$\left(\frac{x}{X_0}\right)_{\text{tot}} = \frac{x_{\text{Si}}}{X_{0,\text{Si}}} + \frac{x_{\text{C}}}{X_{0,\text{C}}}$$

Applying the parameterized formula to this composite material yields the theoretical expectation:

$$\theta_{\text{layer}} = 9.85 \times 10^{-6} \text{ rad}$$



The theta_{MS} histogram specifically isolates the scattering within this single layer. The simulated value is:

$$\theta_{\text{sim}} \approx 1.14 \times 10^{-5} \text{ rad}$$



Exercise 1.4: Expected track curvature

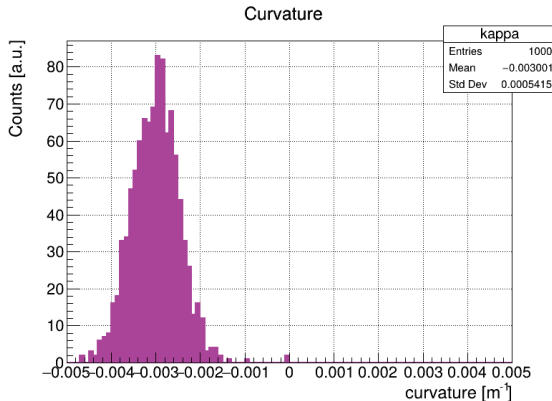
A μ^- with momentum $p = 100 \text{ GeV}/c$ and uniform magnetic field $B = 1 \text{ T}$.

The track curvature $\kappa = q/R$ can be derived:

$$p[\text{GeV}/c] = 0.3 \cdot B[\text{T}] \cdot R[\text{m}]$$

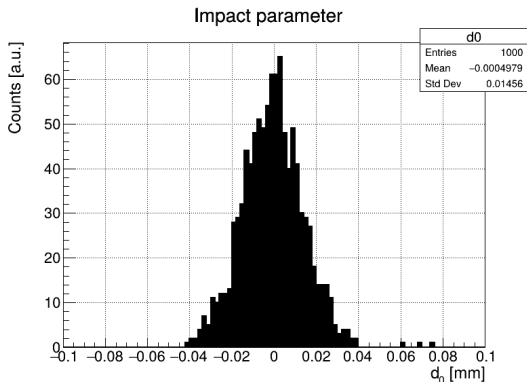
$$\kappa = \frac{-0.3 \cdot B}{p} = -0.003 \text{ m}^{-1}$$

The extracted kappa histogram exhibits a mean value of -0.003 m^{-1} .





Exercise 1.5: Impact parameter resolution (d_0)



The d_0 distribution yields a standard deviation of $\sigma_{\text{sim}} = 14.56 \mu\text{m}$.

The general impact parameter resolution is $\sigma_{d_0} = \sigma_{\text{geom}} \oplus \frac{\sigma_{\text{MS}}}{p \sin^{3/2} \theta}$. For a high-momentum 100 GeV/c muon, MS becomes negligible, leaving $\sigma_{d_0} \approx \sigma_{\text{geom}}$.

Applying the 2-plane approximation using the first ($x_1 = 5 \text{ cm}$) and last ($x_2 = 45 \text{ cm}$) layers, the lever arm is $n = x_2/x_1 = 9$.

Given the intrinsic hit resolution $\sigma_y = 10 \mu\text{m}$:

$$\sigma_{d_0} = \sigma_y \frac{\sqrt{1+n^2}}{n-1} = 11.32 \mu\text{m}$$



Exercise 1.6: Curvature resolution

Using the 3-point sagitta method, the curvature κ is:

$$\kappa \approx \frac{8s}{qL^2}$$

The sagitta is measured using three equidistant detector layers (y_1, y_2, y_3):

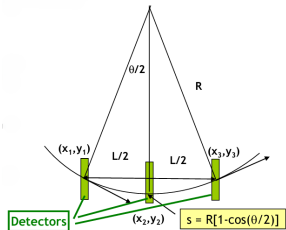
$$s = y_2 - \frac{y_1 + y_3}{2}$$

Assuming an independent spatial resolution σ_y for each layer, the error on the sagitta propagates as:

$$\sigma_s^2 = \sigma_y^2 + \frac{\sigma_y^2}{4} + \frac{\sigma_y^2}{4} = \frac{3}{2}\sigma_y^2$$

So, the geometric resolution limit:

$$\sigma_\kappa = \frac{8}{L^2} \sqrt{\frac{3}{2}} \sigma_y = \frac{\sqrt{96}}{L^2} \sigma_y$$



Source: Prof. Andreazza's Slides

Using $L = 40$ cm (distance from $x_1 = 5$ to $x_3 = 45$ cm) and $\sigma_y = 10$ μ m:

$$\sigma_{\kappa, \text{th}} = \frac{\sqrt{96}}{(0.4 \text{ m})^2} \cdot 10^{-5} \text{ m} = 6.12 \times 10^{-6} \text{ m}^{-1}$$

Recalling the simulated kappa resolution:

$$\sigma_{\kappa, \text{sim}} = 5.42 \times 10^{-6} \text{ m}^{-1}$$



Exercise 2.1: The parabolic approximation error

Exact circular trajectory

A track originating at $(0, R)$ with radius R follows a circle:

$$\begin{aligned} z^2 + (y - R)^2 &= R^2 \\ \implies y^2 - 2Ry + z^2 &= 0 \end{aligned}$$

Solving for the transverse coordinate y (taking the root closest to the origin):

$$\begin{aligned} y_c(z) &= R - \sqrt{R^2 - z^2} \\ y_c(z) &= R \left(1 - \sqrt{1 - \left(\frac{z}{R}\right)^2} \right) \end{aligned}$$

Taylor expansion & error

For high-momentum tracks ($z \ll R$), we apply the binomial expansion $\sqrt{1-x} \approx 1 - \frac{x}{2} - \frac{x^2}{8}$:

$$y_c(z) \approx R \left[1 - \left(1 - \frac{1}{2} \frac{z^2}{R^2} - \frac{1}{8} \frac{z^4}{R^4} \right) \right]$$

$$y_c(z) \approx \frac{z^2}{2R} + \frac{z^4}{8R^3}$$

Tracking algorithms use the parabolic model $y_p(z) = \frac{z^2}{2R}$. The theoretical error is:

$$\Delta y = |y_p - y_c| \approx \frac{z^4}{8R^3}$$



Exercise 2.2: Magnetic field limit

Validity condition

For a valid approximation, the error at the tracker's edge ($z = L = 0.45$ m) must be smaller than the intrinsic hit resolution ($\sigma_y = 10$ μ m).

$$\Delta y(L) \leq \sigma_y \implies \frac{1}{8} \frac{L^4}{R^3} \leq \sigma_y$$

Substituting $R = \frac{p}{0.3B}$:

$$\frac{L^4}{8} \left(\frac{0.3B}{p} \right)^3 \leq \sigma_y$$

$$\implies B^3 \leq \frac{8\sigma_y p^3}{(0.3)^3 L^4}$$

Calculation for a low- p muon

Evaluating for $p = 0.2$ GeV/ c :

$$B \leq \sqrt[3]{\frac{8 \cdot (10^{-5} \text{ m}) \cdot (0.2 \text{ GeV}/c)^3}{0.027 \cdot (0.45 \text{ m})^4}}$$

$$B_{\text{max}} \approx \mathbf{83 \text{ mT}}$$

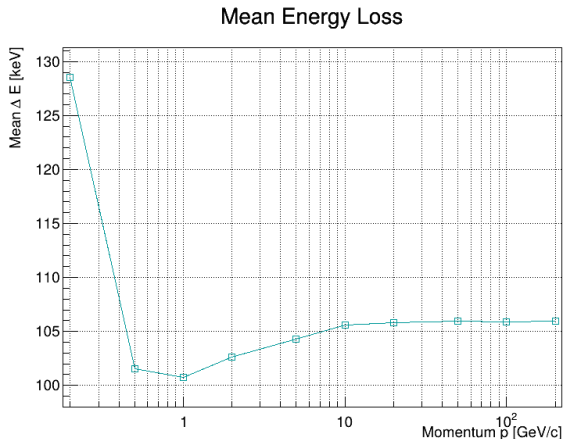
At the nominal field $B = 1$ T, for the same 200 MeV/ c track, the bending radius is $R \approx 0.667$ m. The approximation error becomes $\Delta y \approx 1.73$ cm, completely surpassing the 10 μ m detector resolution.



Exercise 2.3: Mean energy loss vs momentum

The simulated mean energy loss perfectly reproduces the theoretical Bethe-Bloch behavior for muons (200 MeV/c – 200 GeV/c).

- **Low momentum:** The sharp initial drop scales as $1/\beta^2$.
- **MIP regime:** The minimum ionizing region occurs around $p \approx 1 \text{ GeV}/c$.
- **Relativistic rise:** A logarithmic relativistic increase is visible at higher momenta.





Exercise 2.3: Multiple scattering angle

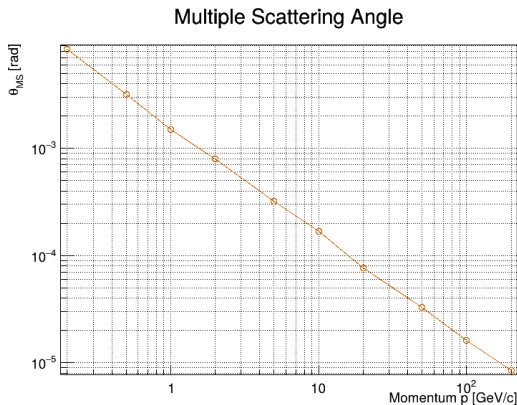
The multiple Coulomb scattering angle is inversely proportional to momentum:

$$\theta_0 \propto \frac{1}{\beta c p} \sqrt{\frac{x}{X_0}}$$

For relativistic particles ($\beta \rightarrow 1$), taking the logarithm yields:

$$\log(\theta_0) = -\log(p) + C$$

As expected, the simulation perfectly follows a straight line with a slope of -1 on the log-log plot.



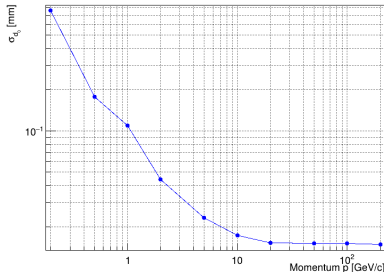


Exercise 2.3: Impact parameter & direction resolutions

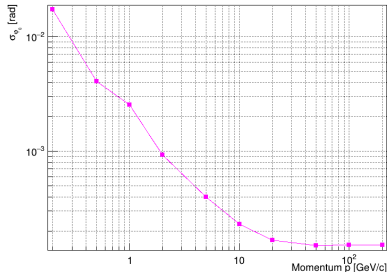
Both σ_{d_0} and σ_{ϕ_0} exhibit the same dual behavior, governed by: $\sigma = A_{\text{geom}} \oplus \frac{B_{\text{MS}}}{p}$

- **MS dominated** ($p < 10 \text{ GeV}/c$): The resolution degrades rapidly as $1/p$, completely dominating the error.
- **Intrinsic limit** ($p > 10 \text{ GeV}/c$): The MS term vanishes, and the resolution asymptotes to the purely geometric detector limit ($\propto \sigma_y$).

Impact Parameter Resolution



Direction Resolution





Exercise 2.3: Curvature resolution (σ_κ)

The curvature resolution also follows the $\sigma = A_{\text{geom}} \oplus \frac{B_{\text{MS}}}{p}$ rule.

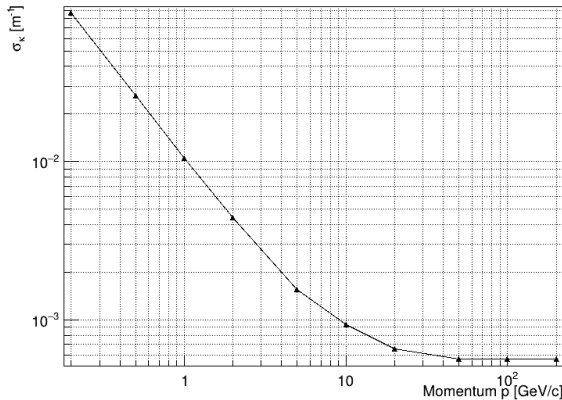
To understand the momentum resolution, we link it to κ . From $p = 0.3B/\kappa$, we propagate the error by differentiating:

$$\left| \frac{dp}{d\kappa} \right| = \frac{0.3B}{\kappa^2} = \frac{p}{\kappa}$$

Therefore, the absolute error on momentum is $\sigma_p = \frac{p}{\kappa} \sigma_\kappa$, which leads to:

$$\frac{\sigma_p}{p} = \frac{\sigma_\kappa}{\kappa}$$

Curvature Resolution





Exercise 2.3: Relative momentum resolution (σ_p/p)

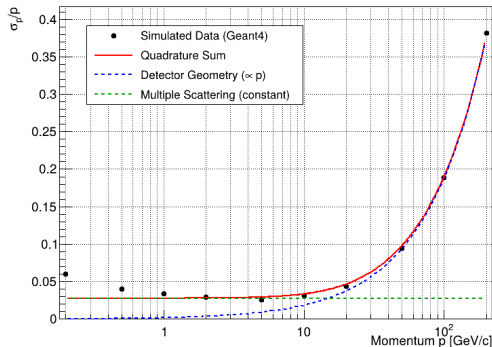
Substituting $\kappa \propto 1/p$, the behavior flips:

$$\frac{\sigma_p}{p} = \sqrt{\underbrace{\left(\frac{A \cdot p}{\text{Geom}}\right)^2}_{\text{Geom}} + \underbrace{B^2}_{\text{MS}}}$$

Parabolic failure at low p

At $p = 0.2 \text{ GeV}/c$, $R = 66.7 \text{ cm}$. The track is no longer a small arc but sweeps a huge angle ($\theta \approx 38^\circ$) inside the tracker, completely breaking the parabolic approximation $\Delta y = \frac{z^4}{8R^3}$.

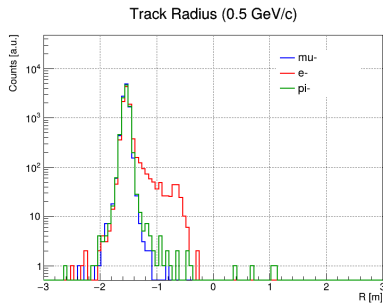
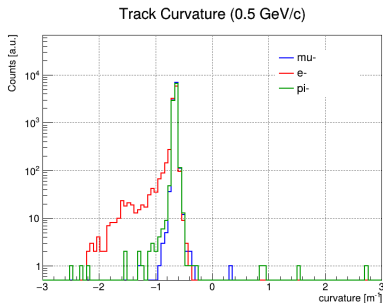
Momentum Resolution Decomposition





Exercise 3.1: 500 MeV/c (κ & R)

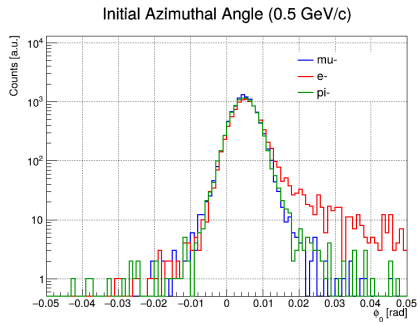
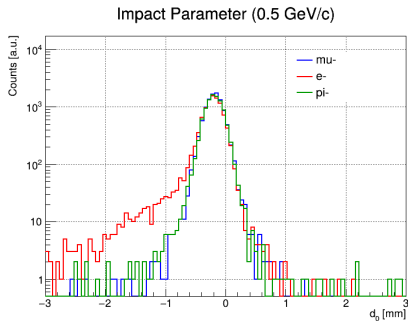
- While μ^- and π^- lose energy steadily via ionization, e^- can do Bremsstrahlung at any point in the tracker.
- A sudden drop in p causes the trajectory to curl tighter ($\kappa \propto 1/p$). The global fit averages this anomaly, creating prominent asymmetric tails.





Exercise 3.1: 500 MeV/c (d_0 & ϕ_0)

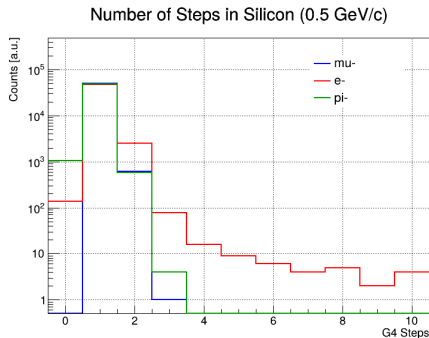
The origin is reconstructed by assuming continuous energy loss. This produces the non-perfectly symmetric distributions observed for d_0 and ϕ_0 .





Exercise 3.1: 500 MeV/c (Number of steps)

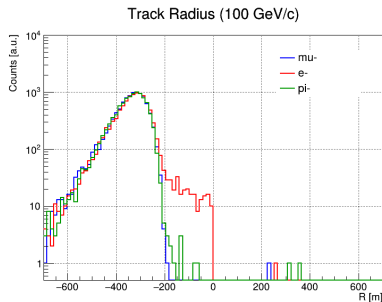
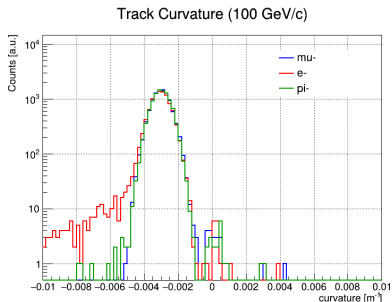
- Geant4 typically simulates a MIP crossing a thin silicon layer in a single step (ionization).
- To generate a secondary Bremsstrahlung photon, Geant4 must interrupt and split the step. The e^- tail is the direct microscopic proof of these discrete events.
- The material is minimal ($\sim 4.5\% X_0$), making hard Bremsstrahlung statistically rare. The logarithmic scale is crucial to reveal this small fraction of non-ideal events against the bulk behavior.





Exercise 3.1: 100 GeV/c (κ & R)

- The e^- still radiates at 100 GeV/c. The continuous loss of momentum preserves the characteristic asymmetric tail.
- The SiTracker is too thin ($300\text{ }\mu\text{m} \ll \lambda_{int}$) to induce hadronic showers. In this case (as well as for 500 MeV particles), pions behave exactly like muons.

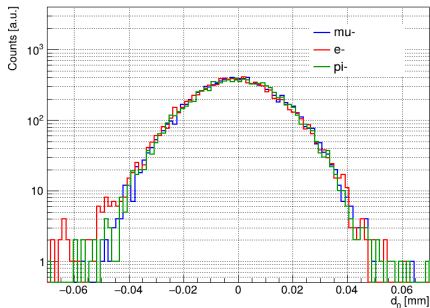




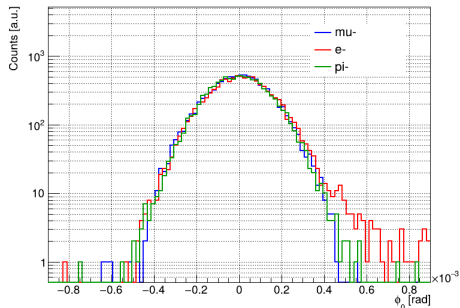
Exercise 3.1: 100 GeV/c (d_0 & ϕ_0)

Despite the radiation shown previously, the tracks are now so rigid that the impact parameter and azimuthal angle are completely unaffected.

Impact Parameter (100 GeV/c)



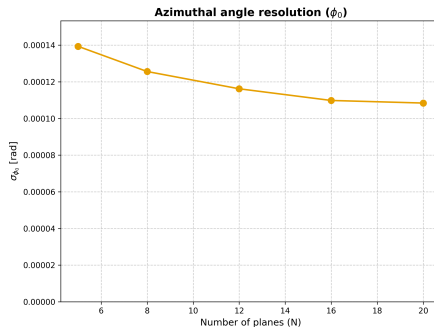
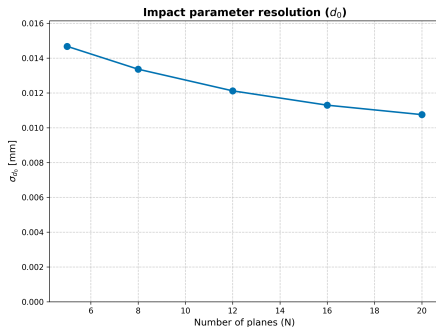
Initial Azimuthal Angle (100 GeV/c)





Exercise 3.2: Local parameters resolution (d_0 & ϕ_0)

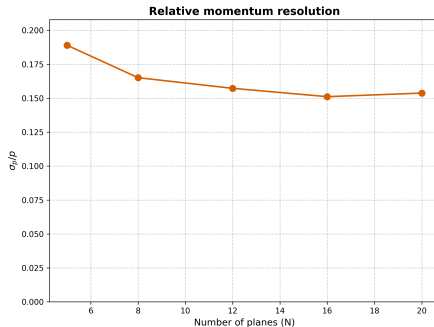
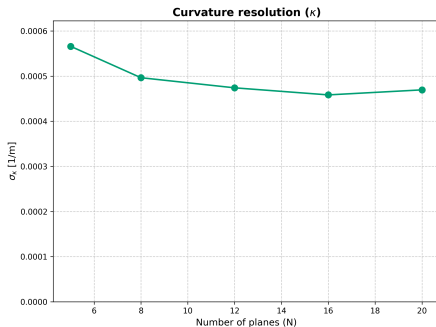
- **Impact parameter (d_0):** Resolution improves monotonically, driven by the increased statistical sampling near the interaction point.
- **Azimuthal angle (ϕ_0):** Exhibits a similar trend, being primarily sensitive to the first few layers before significant material effects accumulate.





Exercise 3.2: Global parameters resolution (κ & σ_p/p)

- **Curvature (κ):** Initially improves statistically, but begins to degrade at $N \sim 20$ planes as the added material introduces significant multiple scattering.
- **Momentum (σ_p/p):** Mirrors the curvature, clearly showing the physical trade-off between higher spatial sampling density and cumulative scattering noise.





Exercise 3.3: Increasing the lever arm

- Doubling the spacing between the 5 planes doubles the total length L (from 40 cm to 80 cm), keeping the material constant.
- Geometric limits predict $\sigma_\kappa \propto 1/L^2$. Conversely, σ_{d_0} with the standard spacing already satisfies $n \gg 1$ implies only a marginal gain with larger L .

Resolution	Standard Spacing	Doubled Spacing	Improvement Ratio
σ_{d_0} [mm]	1.47×10^{-2}	1.17×10^{-2}	~ 1.26
σ_{ϕ_0} [rad]	1.39×10^{-4}	6.49×10^{-5}	~ 2.14
σ_κ [m^{-1}]	5.66×10^{-4}	1.39×10^{-4}	~ 4.07 (Expected ≈ 4)
σ_p/p	1.89×10^{-1}	4.66×10^{-2}	~ 4.065 (Expected ≈ 4)



Exercise 4: Kinematics of $J/\psi \rightarrow \mu^+ \mu^-$ decay

- **Muon kinematics:** Assuming a symmetric decay, the muon momentum p_μ and opening angle ϕ are derived from the invariant mass:

$$p_\mu = \sqrt{(E_{J/\psi}/2)^2 - m_\mu^2} \quad \text{and} \quad \cos \phi = 1 - \frac{m_{J/\psi}^2}{2p_\mu^2}$$

- **Boost and B_s^0 decay:** The Lorentz factor is $\gamma\beta = p_{J/\psi}/m_{J/\psi}$, so the expected decay length is $L_B = \gamma\beta c\tau_B$, with intrinsic $c\tau_B = 450 \mu\text{m}$.
- **Invariant mass:** In the ultrarelativistic approximation, $E_\mu \simeq p_\mu c$ and, for a symmetric decay, $M_{\mu\mu} \simeq 2p_\mu \sin(\phi/2)$. Both cases give $M_{\mu\mu} \simeq 3.09 \text{ GeV}/c^2$, consistent with the J/ψ mass.



Exercise 4: Error propagation model

- **Vertex resolution (σ_{vertex}):** Error propagation from the geometric intersection of the two outgoing tracks in a symmetric decay:

$$\sigma_{vertex} = \frac{\sigma_{d_0}}{\sqrt{2} \tan(\varphi/2)}$$

- **Invariant mass relative errors (σ_M/M):** Propagating relative errors for uncorrelated variables p and ϕ :

$$\frac{\sigma_M}{M} = \sqrt{\left(\frac{\sigma_p}{p}\right)^2 + \left(\frac{\sigma_\phi}{2 \tan\left(\frac{\phi}{2}\right)}\right)^2}$$



Exercise 4: Simulation results

Table: Calculated kinematic values

Parameter	Case A	Case B
p_μ [GeV/c]	1.624	5.233
ϕ	145.0°	34.4°
L_B [mm]	0.145	1.453

Table: Extracted resolutions and derived errors

Uncertainty	Case A	Case B
σ_{d_0} [mm]	0.046	0.021
σ_κ [m ⁻¹]	0.005	0.002
σ_{vertex} [mm]	0.01	0.04
σ_M/M [%]	2.75	3.39



Exercise 4: Discussion and conclusions

- In both simulated regimes, the vertex resolution is strictly smaller than the expected flight length ($\sigma_{vertex} < L_B$). This confirms the detector's capability to separate secondary J/ψ decays from prompt primary vertices.
- The angular contribution to the invariant mass variance ($\sim 10^{-7}$) is completely negligible compared to the momentum measurement error ($\sim 10^{-3}$).
- The resulting mass resolution ($\sim 100 \text{ MeV}/c^2$) completely masks the intrinsic J/ψ natural width of 93 keV.



Silicon Tracker Simulation and Performance Analysis using Geant4

Thank you for listening!
Any questions?



BACKUP



Invariant mass resolution

For a symmetric 2-body decay (e.g., $J/\psi \rightarrow \mu^+ \mu^-$ where $p_1 = p_2 \equiv p_\mu$), the invariant mass simplifies to:

$$M_{\mu\mu} \approx 2p_\mu \sin\left(\frac{\phi}{2}\right)$$

Applying relative error propagation yields a compact formula driven by two main terms:

$$\frac{\sigma_{M_{\mu\mu}}}{M_{\mu\mu}} = \sqrt{\left(\frac{\sigma_{p_\mu}}{p_\mu}\right)^2 + \left(\frac{\sigma_\phi}{2 \tan(\phi/2)}\right)^2}$$

- **Momentum term:** The relative momentum resolution is directly determined by the curvature (κ) resolution of the tracker:

$$\frac{\sigma_{p_\mu}}{p_\mu} = \frac{\sigma_\kappa}{\kappa}$$

- **Angular term ($\phi = \phi_1 - \phi_2$):** Assuming independent and identical angular measurement errors for both tracks ($\sigma_{\phi_1} = \sigma_{\phi_2} \equiv \sigma_{\phi_0}$):

$$\sigma_\phi = \sqrt{\sigma_{\phi_1}^2 + \sigma_{\phi_2}^2} = \sqrt{2}\sigma_{\phi_0}$$



Vertex resolution derivation

Symmetric decay along the x -axis with opening angle φ . Linear track approximation near the origin:

$$y_1 = \tan(\varphi/2)x + d_1$$

$$y_2 = -\tan(\varphi/2)x + d_2$$

Equating the trajectories ($y_1 = y_2$) to find the intersection point along the flight direction:

$$x_v = \frac{d_2 - d_1}{2 \tan(\varphi/2)}$$

Assuming negligible angular uncertainty and independent, identical impact parameter errors ($\sigma_{d_1} = \sigma_{d_2} \equiv \sigma_{d_0}$):

$$\sigma_{x_v}^2 = \left(\frac{\partial x_v}{\partial d_1} \right)^2 \sigma_{d_0}^2 + \left(\frac{\partial x_v}{\partial d_2} \right)^2 \sigma_{d_0}^2$$

Evaluating the partial derivatives directly yields the longitudinal resolution:

$$\sigma_{x_v} = \sqrt{\frac{\sigma_{d_0}^2}{4 \tan^2(\varphi/2)} + \frac{\sigma_{d_0}^2}{4 \tan^2(\varphi/2)}}$$
$$\sigma_{x_v} = \frac{\sigma_{d_0}}{\sqrt{2} \tan(\varphi/2)}$$